lies the quantum bit, or qubit. While a classical bit can exist in one of two definite states—either 0 or 1—a qubit can exist in a state that is both 0 and 1 at the same time, thanks to the unique properties of quantum physics. Imagine a coin spinning in the air: instead of being heads or tails, it is both until it lands. Similarly, a qubit's state is not fixed until it is measured.

Qubits can be realised using various physical systems, such as electrons, photons, or even atoms. For instance, the spin of an electron (up or down) or the polarisation of a photon (horizontal or vertical) can represent the two states of a qubit. By manipulating these systems, scientists can create and control qubits, paving the way for quantum computations.

Comparing classical and quantum computing

The distinction between classical and quantum computing is not merely academic; it has profound implications for what problems can be solved and how efficiently.

Classical computers process information using transistors that represent binary states (0 or 1). They are highly efficient for a wide range of everyday tasks—word processing, web browsing, and even complex simulations. Quantum computers, by contrast, leverage superposition and entanglement to process information in fundamentally new ways. They are ideally suited for solving certain types of problems that are practically insurmountable for classical computers.

The table below highlights the differences between classical and quantum computing.

Quantum gates and circuits: The building blocks

Just as classical computers use logic gates (AND, OR, NOT) as the building blocks of circuits, quantum computers use quantum gates. These gates manipulate qubits through operations that exploit quantum phenomena. Some common quantum gates include:

Basis of difference	Classical computing	Quantum computing
Basic unit of data	Bit – can be either 0 or 1	Qubit – can be 0, 1, or both simultaneously (superposition)
Computation model	Deterministic – follows definite logical operations	Probabilistic – operates using probabilities and quantum states
Processing capability	Sequential or limited parallelism	Massive parallelism due to superposition and entanglement
State representation	Single state at a time	Multiple states at once
Interdependence	Independent bits	Entangled qubits are correlated across distances
Error handling	Uses classical error correction methods	Requires quantum error correction due to fragile qubit states
Algorithms	Classical algorithms (e.g., binary search, sorting)	Quantum algorithms (e.g., Shor's, Grover's) offering exponential or quadratic speedup
Hardware components	Transistors, silicon chips	Quantum processors using superconducting circuits, trapped ions, or photons
Energy consumption	Relatively high due to heat dissipation	Potentially lower for specific tasks, but requires extreme cooling (near absolute zero)
Applications	General-purpose computing: office tasks, internet, AI, databases	Specialised tasks: cryptography, optimisation, quantum simulation, machine learning
Scalability	Mature, scalable with Moore's Law (though slowing)	Emerging, scaling limited by qubit stability and decoherence
Nature of output	Definite and deterministic	Probabilistic; requires measurement and statistical analysis
Computation speed	Linear or polynomial time for most problems	Exponential speedup for certain problem classes